STT 861: Theory of Probability & Statistics I

Lecture 15: Poisson, Geometric & Negative Binomial

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Reference: Casella & Berger Ch. 3.2

Discrete distributions: overview

Definition (CB)

X has a discrete distribution if its range (the sample space) is countable. Usually the outcomes are integer-valued.

Families in this section (book order)

- 1. Discrete uniform 2. Hypergeometric 3. Binomial (Bernoulli trials)
- 4. Poisson 5. Negative binomial 6. Geometric

For each we record the pmf, mean, variance, mgf, a story, and a worked example from the book.

Discrete uniform distribution

Definition (CB 3.2.1)

$X \sim$ discrete uniform$(1, N)$ if $P(X = x \mid N) = \dfrac{1}{N}, \quad x = 1, 2, \ldots, N$, where N is a specified integer: equal mass on each outcome.

Mean and variance (book derivation)

Use the identities (provable by induction) $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ and $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$. Then

$$\mathbb{E}X = \sum_{x=1}^{N} \frac{x}{N} = \frac{N+1}{2}, \quad \mathbb{E}X^2 = \frac{(N+1)(2N+1)}{6}, \quad \operatorname{Var} X = \frac{(N+1)(2N+1)}{6} - \Big(\frac{N+1}{2}\Big)^2 = \frac{(N+1)(N-1)}{12}.$$

Generalization and a check

Any integer range $N_0, N_0 + 1, \ldots, N_1$ has pmf $P(X = x \mid N_0, N_1) = 1/(N_1 - N_0 + 1)$. Fair die ($N = 6$): $\mathbb{E}X = 3.5$, $\operatorname{Var} X = 35/12 \approx 2.92$.

Hypergeometric distribution

Urn model (CB 3.2.2)

An urn holds N balls: M red, $N-M$ green. Draw K balls all at once (without replacement). With $X =$ number of red balls,

$$P(X = x \mid N, M, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}},$$

with range $M-(N-K) \le x \le M$ (and $0 \le x \le K$).

Mean and variance

$$\mathbb{E}X = \frac{KM}{N}, \quad \operatorname{Var} X = \frac{KM}{N} \cdot \frac{(N-M)(N-K)}{N(N-1)}.$$

Proof idea (mean): use $x\binom{M}{x} = M\binom{M-1}{x-1}$ and $\binom{N}{K} = \frac{N}{K}\binom{N-1}{K-1}$. The remaining sum is the total probability of a hypergeometric$(N-1, M-1, K-1)$ pmf, so it equals 1.

Remark from the book

It is not even trivial to verify $\sum_x P(X = x) = 1$: finite populations (finite N) make statistics difficult.

Example 3.2.1: acceptance sampling

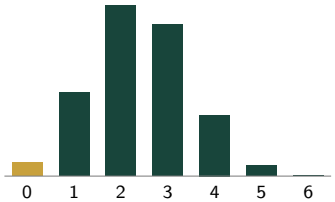
A retailer receives a lot of $N = 25$ machine parts; M are defective. Sample $K = 10$ and find none defective. How likely is that if the lot has $M = 6$ defectives?

$$P(X = 0) = \frac{\binom{6}{0}\binom{19}{10}}{\binom{25}{10}} = .028.$$

Conclusion (book)

The observed event is quite unlikely if there are 6 (or more!) defectives in the lot.

Here $\mathbb{E}X = KM/N = 10 \cdot 6/25 = 2.4$: we would *expect* about 2–3 defectives in the sample.



pmf of X for $(N, M, K) = (25, 6, 10)$; the gold bar is $P(X = 0)$.

Bernoulli and binomial distributions

Bernoulli(p) (CB 3.2.3)

$X = 1$ with probability p ("success"), $X = 0$ with probability $1 - p$, $0 \le p \le 1$. $\mathbb{E}X = p$, $\operatorname{Var} X = (1-p)^2 p + (0-p)^2(1-p) = p(1-p)$.

Binomial(n, p): derivation

n independent Bernoulli trials; $A_i = \{X_i = 1\}$ and $Y =$ total number of successes. One particular sequence with exactly y successes has probability $p^y(1-p)^{n-y}$, by independence. There are $\binom{n}{y}$ such orderings, so

$$P(Y = y \mid n, p) = \binom{n}{y} p^y (1-p)^{n-y}, \qquad y = 0, 1, \ldots, n.$$

Equivalently $Y = X_1 + \cdots + X_n$ with $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.

Book's examples of Bernoulli trials: coin tossing, roulette ($X = 1$ if red), election polls ($X = 1$ if candidate A gets a vote), disease incidence ($p =$ probability a random person is infected).

Theorem 3.2.2 (Binomial Theorem) and binomial moments

Theorem 3.2.2 (Binomial Theorem)

For any real numbers x and y and integer $n > 0$,

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}. \tag{3.2.4}$$

Proof idea: write $(x+y)^n = (x+y)(x+y)\cdots(x+y)$. From each factor choose x or y and multiply the n choices. The number of terms in which x appears exactly i times is $\binom{n}{i}$.

Consequences

$x = p,\ y = 1-p$: $1 = \sum_{y=0}^{n} \binom{n}{y} p^y (1-p)^{n-y}$, so the binomial pmf sums to 1.

$x = y = 1$: $2^n = \sum_{i=0}^{n} \binom{n}{i}$.

Moments (Examples 2.2.3, 2.3.5, 2.3.9)

$\mathbb{E}X = np$, $\operatorname{Var} X = np(1-p)$,
$M_X(t) = \left[pe^t + (1-p)\right]^n$.

Example 3.2.3: dice probabilities (de Méré's problem)

Game 1: at least one 6 in four rolls of a fair die.
$X \sim \text{Bin}(4, \frac{1}{6})$:

$$P(X > 0) = 1 - P(X = 0) = 1 - \left(\frac{5}{6}\right)^4 = .518.$$

Game 2: at least one double 6 in 24 throws of a pair of dice.
$p = \frac{1}{36}$, $Y \sim \text{Bin}(24, \frac{1}{36})$:

$$P(Y > 0) = 1 - \left(\frac{35}{36}\right)^{24} = .491.$$



History (from the book)

Pascal did this calculation in the eighteenth century at the request of the gambler de Méré, who thought both events were equally likely. He began to believe he was wrong when he started losing money on the second bet.

Poisson distribution

Definition (CB 3.2.5)

X takes values in $\{0, 1, 2, \dots\}$ and has a Poisson(λ) distribution if

$$P(X = x \mid \lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$\lambda > 0$ is the intensity parameter. The sum is 1 by the Taylor series $e^y = \sum_{x \geq 0} y^x / x!$.

Moments and mgf

$\mathbb{E}X = \lambda$, $\operatorname{Var} X = \lambda$ (mean = variance),

$M_X(t) = e^{\lambda(e^t - 1)}$.

Recursion (3.2.6):

$P(X = x) = \frac{\lambda}{x} P(X = x - 1)$.

When is it used? (book)

Waiting for occurrences: counts in a time interval (customers entering a bank, bus arrivals) when, for small intervals, the probability of an arrival is proportional to the length of waiting time.
Spatial distributions: bomb hits in an area, fish in a lake.

Example 3.2.4: waiting time (telephone operator)

An operator handles on average five calls every 3 minutes. Let X = number of calls in one minute, so $X \sim \text{Poisson}(\lambda)$ with $\lambda = \mathbb{E}X = \frac{5}{3}$.

Probabilities

$$P(\text{no calls in the next minute}) = P(X = 0) = \frac{e^{-5/3}(5/3)^0}{0!} = e^{-5/3} = .189,$$

$$P(\text{at least two calls}) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-5/3}\left(1 + \tfrac{5}{3}\right) = .496.$$

Fast computation: recursion (3.2.6)

Start from $P(X = 0) = e^{-5/3}$; then $P(X = 1) = \frac{5/3}{1} P(X = 0)$, $P(X = 2) = \frac{5/3}{2} P(X = 1)$, and so on, with no factorials needed.

Poisson approximation to the binomial

Why it works (book argument)

If $Y \sim \mathrm{Bin}(n, p)$ then (3.2.7)

$$P(Y = y) = \frac{n - y + 1}{y} \frac{p}{1 - p} P(Y = y - 1)$$

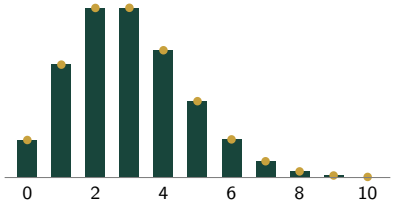
$$\approx \frac{\lambda}{y} P(Y = y - 1)$$

with $\lambda = np$ and p small (the terms $p(y-1)$ and py are negligible): the *Poisson recursion*. Also $P(Y = 0) = (1 - \lambda/n)^n \to e^{-\lambda} = P(X = 0)$. Good when n is large and p is small.

Example 3.2.5 (typesetter). One error per 500 words on average; 5 pages = 1500 words. $X \sim \mathrm{Bin}(1500, \frac{1}{500})$, and

$$P(X \le 2) = .4230 \text{ (exact)},$$

$$\text{Poisson}(\lambda{=}3) : e^{-3}\left(1 + 3 + \tfrac{9}{2}\right) = .4232.$$



Bars: Poisson(3). Gold dots: Bin(1500, 1/500). They are visually indistinguishable.

Negative binomial distribution

Trial of the rth success (CB 3.2.9)

In independent Bernoulli(p) trials, let $X =$ the trial at which the rth success occurs (r fixed):

$$P(X = x \mid r, p) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, \ldots$$

Derivation: exactly $r-1$ successes in the first $x-1$ trials (binomial), then a success on trial x (probability p).

Book's default form (3.2.10): $Y =$ failures before the rth success

$Y = X - r$ and

$$P(Y = y) = \binom{r+y-1}{y} p^r (1-p)^y, \quad y = 0, 1, 2, \ldots; \qquad \mathbb{E}Y = \frac{r(1-p)}{p}, \quad \text{Var } Y = \frac{r(1-p)}{p^2}.$$

The name: $\binom{r+y-1}{y} = (-1)^y \binom{-r}{y}$ (negative-integer binomial coefficient).

Negative binomial: reparametrization, Poisson limit, Example 3.2.6

Reparametrize by the mean

Let $\mu = r(1-p)/p$. Then $\mathbb{E}Y = \mu$ and

$$\operatorname{Var} Y = \mu + \frac{\mu^2}{r} :$$

the variance is a quadratic function of the mean.

Poisson as a limiting case

If $r \to \infty$ and $p \to 1$ with $r(1-p) \to \lambda$, then $\mathbb{E}Y \to \lambda$ and $\operatorname{Var} Y \to \lambda$, the Poisson mean and variance. In fact $\mathrm{NB}(r,p) \to \mathrm{Poisson}(\lambda)$ (probabilities converge; Exercise 3.15).

Example 3.2.6: inverse binomial sampling

Fruit flies: a proportion p have vestigial wings. Sample until $r = 100$ such flies are found. The number sampled X is negative binomial, so

$$P(X \geq N) = \sum_{x=N}^{\infty} \binom{x-1}{99} p^{100}(1-p)^{x-100}.$$

For given p and N this tells us how many flies we are likely to examine.

Like the Poisson, it models *waiting for an occurrence*: here we wait for a specified number of successes.

Geometric distribution and the memoryless property

Geometric(p) = negative binomial with $r = 1$

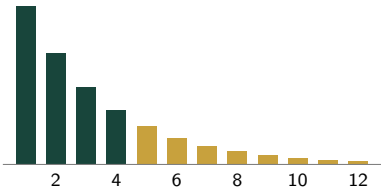
$P(X = x \mid p) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$ (X = trial of the first success).
$\sum_x P(X = x) = 1$ by the geometric series.
$\mathbb{E}X = \dfrac{1}{p}$, $\quad \operatorname{Var} X = \dfrac{1-p}{p^2}$.

Memoryless property (CB 3.2.11)

For integers $s > t$: $P(X > s \mid X > t) = P(X > s - t)$.

Proof: $P(X > n) = (1-p)^n$ (no success in n trials, (3.2.12)), so

$$P(X > s \mid X > t) = \frac{P(X > s)}{P(X > t)} = (1-p)^{s-t}.$$



Geometric(0.3) pmf. The gold tail $\{X > 4\}$ is a rescaled copy of the whole pmf.

Example 3.2.7: failure times

Light bulb

The probability a bulb fails on any given day is .001. Let $X =$ day of failure, $X \sim \text{Geometric}(.001)$:

$$P(X > 30) = \sum_{x=31}^{\infty} .001(1 - .001)^{x-1} = (.999)^{30} = .970.$$

Caveat (book)

The memoryless property is a very special "lack of aging" property. The geometric distribution is not appropriate for lifetimes whose failure probability is expected to increase with time. Other distributions model aging (Barlow and Proschan, 1975).

Summary table of discrete families

Family	pmf	Mean	Variance	mgf
Discrete unif.$(1, N)$	$\frac{1}{N},\ x = 1..N$	$\frac{N+1}{2}$	$\frac{(N+1)(N-1)}{12}$	$\frac{1}{N}\sum_{k=1}^{N} e^{tk}$
Hypergeom.(N, M, K)	$\binom{M}{x}\binom{N-M}{K-x}/\binom{N}{K}$	$\frac{KM}{N}$	$\frac{KM}{N}\frac{(N-M)(N-K)}{N(N-1)}$	—
Binomial(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$[pe^t + 1 - p]^n$
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	λ	λ	$e^{\lambda(e^t-1)}$
Neg. binomial(r, p) (Y failures)	$\binom{r+y-1}{y}p^r(1-p)^y$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{p}{1-(1-p)e^t}\right)^r$
Geometric(p) (X trial)	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$

The negative binomial and geometric mgfs are defined for $(1-p)e^t < 1$.

Activity 1 — Discrete families by hand

Activity 1

(a) Fair die, $X \sim$ discrete uniform$(1, 6)$. Use the book's formulas for $\mathbb{E}X$ and $\operatorname{Var} X$.

(b) An urn holds 10 balls, 4 red. Draw 3 at once. Find $P(X = 2)$ for $X =$ number of red balls, and $\mathbb{E}X$.

(c) Toss a fair coin 5 times. Find P(at least two heads).

(d) A mailbox receives on average 12 emails per hour. Find the probability of no email in a 10-minute window, and of at least three.

Activity 1— Solutions

(a) $\mathbb{E}X = \frac{6+1}{2} = 3.5$, $\operatorname{Var} X = \frac{(6+1)(6-1)}{12} = \frac{35}{12} \approx 2.92$.

(b) Hypergeometric($N = 10, M = 4, K = 3$): $P(X = 2) = \dfrac{\binom{4}{2}\binom{6}{1}}{\binom{10}{3}} = \dfrac{36}{120} = .30$; $\mathbb{E}X = \dfrac{KM}{N} = \dfrac{3 \cdot 4}{10} = 1.2$.

(c) $X \sim \mathrm{Bin}(5, \frac{1}{2})$: $P(X \geq 2) = 1 - P(0) - P(1) = 1 - \dfrac{1+5}{32} = \dfrac{13}{16} = .8125$.

(d) $\lambda = 12 \cdot \frac{10}{60} = 2$ per 10 minutes. $P(X = 0) = e^{-2} = .135$;
$P(X \geq 3) = 1 - e^{-2}(1 + 2 + 2) = 1 - 5e^{-2} = .323$.

Activity 2 — Waiting times and approximations (12 min, groups of 3)

Activity 2

(a) A component fails on each day independently with probability .02. Let X be the day of the first failure. Name the distribution, find $\mathbb{E}X$ and $P(X > 50)$.

(b) Given the component has survived 20 days, what is the probability it survives at least 30 more? Which property did you use?

(c) Items are defective independently with probability .1. Sample until the 3rd defective is found; let Y be the number of non-defective items seen. Name the distribution of Y, find $\mathbb{E}Y$ and $P(Y = 2)$.

(d) $X \sim \mathrm{Bin}(200, .01)$. Compare the exact $P(X \leq 1)$ with its Poisson approximation.

Activity 2 — Solutions

(a) Geometric(.02): $\mathbb{E}X = 1/p = 50$; $P(X > 50) = (.98)^{50} = .364$.

(b) $P(X > 50 \mid X > 20) = P(X > 30) = (.98)^{30} = .545$, by the memoryless property (3.2.11).

(c) Negative binomial($r = 3, p = .1$) (failures form, 3.2.10): $P(Y = y) = \binom{y+2}{y}(.1)^3(.9)^y$. $\mathbb{E}Y = \dfrac{3(.9)}{.1} = 27$; $P(Y = 2) = \binom{4}{2}(.1)^3(.9)^2 = 6(.001)(.81) = .00486$.

(d) Exact: $(.99)^{200} + 200(.01)(.99)^{199} = .1340 + .2707 = .4046$. Poisson with $\lambda = np = 2$: $e^{-2}(1 + 2) = .4060$. The approximation is accurate to about two decimals.

Summary and next time

Take-aways

- A parametric family $\{f(x \mid \theta) : \theta \in \Theta\}$ turns an unknown distribution into an unknown *number(s)*.
- Parameters control location, scale and shape (e.g. p, μ, σ).
- We want parameters to be identifiable.
- Ch. 2 tools give $\mathbb{E}X$, $\operatorname{Var} X$, $M_X(t)$ as functions of θ.
- Discrete families (): uniform, hypergeometric, binomial, Poisson, negative binomial, geometric; connected by sums, limits and waiting-time stories.

Next lecture

§3.3 Continuous families: uniform, gamma, normal, beta, Cauchy, . . .

Reading: CB §3.3.